

Glossary

Terms defined elsewhere in the glossary are in *italics*.

alchemy

An ancient branch of *chemistry*, which aimed to change ordinary metals into gold.

alloy

A material made by mixing a metal with small amounts of other metals or nonmetals.

antibiotic

A medical drug that kills or slows down the growth of *bacteria*.

antiseptic

A medical drug that kills disease-causing *microbes*. Antiseptics may be applied to the skin to prevent infection.

astronomy

The study of objects in space. An astronomer is a scientist who studies objects in space.

atom

The smallest part of an *element* that has the characteristics of that *element*.

bacteria

A group of single-celled *microbes*, some of which cause disease.

battery

A portable *electricity* supply that stores electric charge using *chemicals*.

biology

A branch of science concerned with living organisms. A biologist is a scientist who studies living things.

black hole

An object in space with a *gravity* so strong that no *matter* or light can escape it.

boiling point

The temperature at which a liquid changes into gas.

bonds

The attraction between *atoms* or groups of *atoms* that holds them together in a *molecule*.

breeding

The mating of two animals to produce offspring.

buoyancy

The upward *force* on an object in a liquid, caused by the water pressure underneath it.

carbohydrate

A *chemical compound*, found in starchy foods such as rice and bread, which gives us *energy*.

cell

The basic unit from which all living organisms are made.

chemical

A substance made from *elements* or *compounds*.

chemistry

A branch of science concerned with the composition of *chemicals* and how they react with each other. A chemist is a scientist who studies *chemicals* and their reactions.

circuit

A path along which *electricity* flows around. All electrical and electronic things have circuits inside them.

climate change

Long-term changes in Earth's weather patterns, resulting from global environmental variations or human activity.

cloning

The process of creating an organism from a body cell of another organism, so they are genetically identical.

combustion

A *chemical* reaction in which a fuel, such as wood or coal, burns with oxygen from the air to release heat *energy*.

compound

A *chemical* made by combining the *atoms* or *molecules* of two or more different *elements*.

condensation

The change of gas or vapor into a liquid.

conservation

The preservation of any process, object, or life.

continent

One of Earth's large land masses, such as Africa.

crankshaft

A rod in a car's *engine* that changes the up and down motion of a *piston* into a rotating motion that turns the car's wheels.

diode

An electronic component that allows an electric current to flow through a *circuit* in only one direction.

dissection

Cutting open of a dead body to study its internal structure.

DNA

Deoxyribonucleic acid. The *chemical* inside chromosomes that lets parents pass genetic information on to their offspring.

electricity

A type of *energy* caused by *electrons* inside *atoms*. Static electricity is made by *electrons* building up in one place, while current electricity happens when *electrons* move around.

electrode

An electrical contact, made from a conductor, that connects the main part of a *circuit* to something outside it, such as the *chemicals* in a *battery*.

electromagnet

A magnet that produces a *magnetic field* because of *electricity*.

electron

A subatomic particle with a negative charge found around an *atom's nucleus*.

element

A basic building block of *matter* made from identical *atoms*.

endangered

A *species* of plant or animal that is at risk of getting *extinct*.

energy

A property of an object that allows it to do something now or in the future. Types of energy include kinetic energy (movement energy) and potential energy (stored energy).

engine

A mechanical device that provides power.

enzyme

A substance that living things use to speed up *chemical* reactions inside them.

evaporation

The change of a liquid into a gas or vapor.

evolution

The process by which *species* change over many generations.

extinct

A *species* that has completely died out.

filament

The part of a light bulb that glows when an electric current flows through it.

food chain

A series of organisms, each of which are consumed by the next.

force

A pushing or pulling action that changes an object's speed, direction of movement, or shape.

fossil

Remains of plants and animals that have been preserved in Earth's crust, or outer layer.

freezing point

The temperature at which a liquid turns into a solid.

frequency

A measurement of how often a wave of *energy* moves up and down.

friction

The rubbing *force* between two things that move past one another. Friction slows things down and generates heat.